

LUCA MALAGUTTI

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🔗 [LucaMalagutti](https://github.com/LucaMalagutti)

Work Experience

Senior Applied Scientist

Apr. 2024 – Present

Thomson Reuters

Zug, Switzerland

- Promoted three times in two years, from Applied Research Intern to Applied Scientist I, II, and Senior Applied Scientist I, in recognition of technical leadership and successful productionization of AI news systems.
- Owned end-to-end technical roadmap for AI-powered news drafting system, productionizing AI features (article generation, urgent updates, spot stories) that process thousands of news articles monthly.
- Architected a scalable LLM data pipeline for financial news, automating the transformation of unstructured releases into 30,000+ alerts per month.
- Implemented production monitoring and model evaluation infrastructure for LLM-based systems, ensuring reliability across multiple deployed features and establishing performance baselines.
- Developed and deployed transformer-based news relevance classification model for financial alerting systems, achieving stakeholder-defined accuracy targets.
- Built LLM-as-Judge evaluation framework for RAG system quality assessment, establishing automated evaluation pipelines that reduced manual QA overhead.
- Led cross-functional technical integration across 4 AI workstreams, serving as primary technical contact between editorial and engineering teams.

Research Assistant

Oct. 2023 – Mar. 2024

ETH Zurich

Zurich, Switzerland

- Developed experimental frameworks and evaluation tools for tokenization and sampling research under Prof. Ryan Cotterell, contributing to papers published at ICLR, ACL, EMNLP, and NAACL (100+ citations).

Data Science Intern

Mar. 2022 – Sep. 2022

SEI Novus

Zurich, Switzerland

- Built and deployed from scratch a complete NLP pipeline for automatic data extraction from financial documents.

Education

MSc in Computer Science

Sep. 2020 – Aug. 2023

ETH Zurich

Zurich, Switzerland

- Teaching Assistant for the Natural Language Processing & Large Language Models courses. Delivered lectures and designed graded assignments and exam questions.
- Selected courses: Natural Language Processing, Advanced Machine Learning, Machine Perception, Big Data.

BSc in Computer Engineering

Sep. 2017 – Jul. 2020

Politecnico di Milano

Milan, Italy

- Final grade: 110/110 - Awarded Best Freshmen Award and merit-based tuition Scholarships.
- Selected courses: Probability and Statistics, Linear Algebra, Algorithms and Principles of Computer Science.

Selected Publications

- Juan Luis Gastaldi, John Terilla, **Luca Malagutti**, Brian DuSell, Tim Vieira, and Ryan Cotterell. "The Foundations of Tokenization: Statistical and Computational Concerns." *ICLR 2025*
- **Luca Malagutti**, Andrius Buinovskij, Anej Svete, Clara Meister, Afra Amini, and Ryan Cotterell. "The Role of n -gram Smoothing in the Age of Neural Networks." *NAACL 2024*
- Clara Meister, Tiago Pimentel, **Luca Malagutti**, Ethan Wilcox, and Ryan Cotterell. "On the Efficacy of Sampling Adapters." *ACL 2023*

Skills & Interests

Programming Languages

ML & NLP

LLMs

MLOps

Languages

Endurance sports

Python, Java, C, SQL, Bash, L^AT_EX

PyTorch, HuggingFace Transformers, vLLM, spaCy, scikit-learn, pandas

Fine-Tuning, LoRA, Prompt Engineering, Evaluation, RAG, Agents

AWS, Docker, Git, CI/CD, Experiment Tracking, Data Pipeline Design

Italian (native), English (professional proficiency), German (B1 level)

Completed full Ironman, 3× Half Ironman, and 3 marathons (PB 2h49m)